MIKHIL GEORGE EDATTEL

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EDUCATION

University of Notre Dame *August 2024 - May 2028*
B.S. in Mechanical Engineering ***GPA:*** *3.73/4.00*
Concentration: Thermal & Fluid Sciences, Computational Engineering
Relevant Coursework: Solid Mechanics, Thermodynamics; Computational Methods, Fluid Mechanics, Differential Eqns (in progress)

Honors: *Certified SolidWorks Associate (CSWA) | President's Circle Mentorship Scholar | Dean's List*

University of Notre Dame London – *Summer Engineering Study Abroad* *May 2025 – June 2025*

EXPERIENCE

Cooper Standard, *CAE Engineer Intern* *Northville, MI | June 2026 – August 2026*

- Achieved a 10 N compression load deflection (CLD) target for a Ford F-150 Lightning outer-belt seal by iterating geometry in MSC Marc/Mentat using nonlinear 2D FEA across +/-2 mm door variation and +2.5 mm overslam conditions
- Determined optimal flange thickness for a Toyota truck sealing application by comparing peak insertion and extraction forces across 0.95, 1.50, and 2.75 mm thicknesses using FEA and documenting findings in a technical report
- Evaluated GM Silverado glass-run seal stability using 3D nonlinear contact FEA in MSC Marc/Mentat, modeling Santoprene and Celloprene installation and predicting buckling, rippling, and bending under overcompression

Domer Rover - Mars Rover Design Team, *Project Manager* *Notre Dame, IN | September 2024 – Present*

- Direct development of a University Rover Challenge rover for a 30+ member team, coordinating mechanical, electrical, autonomy, and communications work around mission requirements and subsystem dependencies
- Lead design reviews and evaluate technical trade-offs for arm and astrobiology modules, resolving conflicting subsystem needs and selecting configurations compatible with the integrated rover
- Redesigned central housing unit from acrylic to aluminum to integrate electronics, computing, and power systems while enabling interchangeable arm and science modules; verified support of a 70 lb. static load through physical testing
- Conducted center-of-mass analysis to guide housing placement relative to the suspension and tested rover mobility across sand, gravel, and rock to identify needed mechanical improvements

RESEARCH & PROJECTS

Hypersonic Systems Lab, *Research Assistant* *Notre Dame, IN | August 2026 – Present*

- Developing an in-house CFD solver in MATLAB and C++ using finite volume methods to simulate compressible flows and shock waves, targeting replacement of commercial ANSYS Fluent workflows for research
- Studying numerical methods for hyperbolic conservation laws through LeVeque's *Finite Volume Methods for Hyperbolic Problems*, building a foundation in shock-capturing schemes and numerical discretization
- Deriving governing equations from conservation of mass, momentum, and energy and reviewing CFD literature to investigate computational approaches for high-speed flow simulation

H.B. Chang Lab, *Research Assistant* *Notre Dame, IN | August 2025 – May 2026*

- Miniaturized a benchtop rotary nanofabrication system into a 500 g handheld device by redesigning solution delivery, airflow mechanisms, and power systems
- Resolved rotor instability at 10,000 RPM by redesigning the turbine-spinneret coupling in SolidWorks, correcting alignment, and validating performance across 20+ experimental runs
- Designed an annular air-jet ring to guide spun fibers toward a collection plate, achieving 95% collection efficiency in controlled deposition trials

Servo-Controlled Marble Maze *March 2026 – May 2026*

- Led mechanical design and build planning for a two-axis marble maze, selecting a nested gimbal architecture to achieve ±20 degrees of tilt on both axes
- Developed SolidWorks models and dimensioned manufacturing drawings, then fabricated and assembled a laser-cut plywood frame, a 3D-printed maze, and dual-servo actuation hardware into a functioning prototype

SKILLS

Simulation:	MSC Marc/Mentat · Ansys Fluent · MATLAB · Nonlinear Contact Analysis · Control Systems
CAD & Design:	SolidWorks (CSWA) · Siemens NX · Engineering Drawings · Dimensional Tolerancing
Programming:	C/C++ · Python · Arduino · ROS2 · Numerical Methods
Manufacturing:	Additive Manufacturing · CNC · Lathe · Laser Cutting · MIG Welding · Waterjet Cutting · Soldering